

< Previous







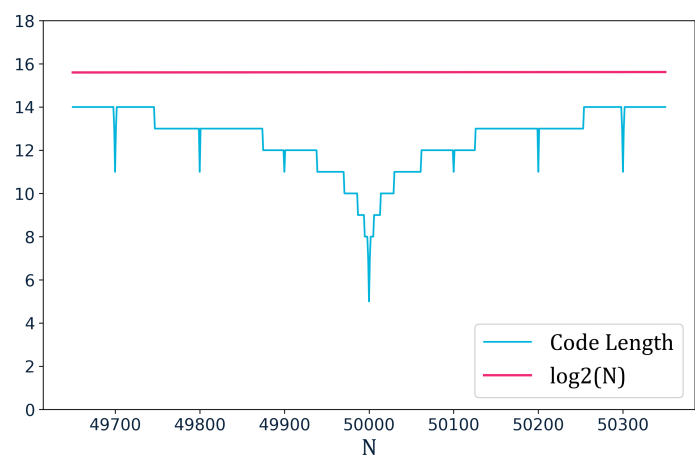
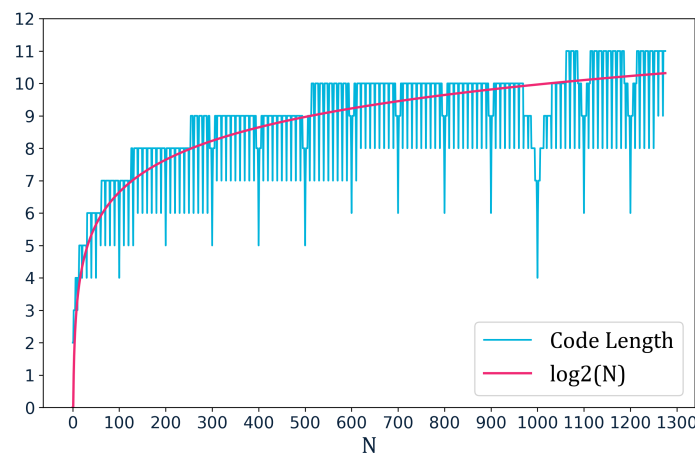
Next >

Complexity is "continuous"

 Bookmark this page

Problem: Does the complexity of integers vary erratically or smoothly?

Thanks to the previous coding method, round numbers and their neighbours receive shorter codes than other numbers of the same magnitude. The two plots below show the complexity of each number n on the x -axis. They have been drawn using the above compact integer coding method, implemented in [a Python program](#). The purple curve indicates $\log_2(n)$ for reference. On a large scale (first image), we can see that the compact code length roughly evolves like $\log(n)$. Note that the complexity cannot exceed $\log_2(n)$ by more than one bit. We can observe the round-number effect: complexity drops for multiples of 10, and even more so for multiples of 100 and of 1000. A zoom around 50000 (second image) shows that $C(n)$ is roughly "continuous".



To capture the fact that the complexity of integers is a "quasi-continuous" function, we may write that for any integers n and h :

$$|C(n+h) - C(n)| \leq f(|h|) + O(1)$$

where f is an increasing function of $|h|$ such that $f(0) = 0$ and $O(1)$ is a constant.

Continuity of complexity

1/1 point (ungraded)

Which function f among the following is the most appropriate to capture the "quasi-continuity" of complexity?

☐ $|h|$

☐ $2 \times |h|$

☒ $\log_2 (1 + |h|)$

☐ $\log_2 (1 + \log_2 (1 + |h|))$

☐ $\exp(|h|) - 1$



Answer

Correct:

Let's consider that $h > 0$. The description of n can be used to describe $n + h$. As a result, $C(n + h) \leq C(n) + C(h)$.

Similarly, $U(n) \leq U(n + n) + U(-n)$, so $U(n + n) - U(n) \geq -U(-n)$. Since $C(-h)$ is close to $C(h)$, an upper bound of $|C(n + h) - C(n)|$ is therefore $C(h)$, which is roughly $\log_2(1 + h)$.

Submit

You have used 1 of 1 attempt

✔ Correct (1/1 point)

< Previous

Next >



edX

- About
- Affiliates
- edX for Business
- Open edX
- Careers
- News

Legal

- Terms of Service & Honor Code
- Privacy Policy
- Accessibility Policy
- Trademark Policy
- Sitemap
- Cookie Policy
- Your Privacy Choices

Connect

- Idea Hub
- Contact Us
- Help Center
- Security
- Media Kit

